SESSION 5

UML Use Case Modeling

AI Career Guidance & Skill Recommendation System

Project	AI Career Guidance & Skill Recommendation System
Deliverable	Use Case Diagram and Narratives
Industry Mapping	Requirement Modeling
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Repository	[Add GitHub repository link]

Topics Covered in This Session

- Use case identification
- Use case diagram creation
- Use case description templates
- Tools: StarUML / Draw.io

1. Actors

Actor	Type	Role in the system
Student	Primary	Uploads resume, views recommendations, gaps, roadmap, courses, uses chatbot and dashboard
Admin	Primary	Manages careers, skills, courses and users; monitors analytics
Claude AI	Secondary (external system)	Analyses resumes, explains recommendations, generates roadmaps and answers chatbot queries

2. Use Case Identification

ID	Use case	Primary actor	Source (user story / FR)
UC-01	Upload Resume (includes Analyze Resume)	Student	US-02, US-03 / FR-06 to FR-10
UC-02	View Career Recommendations (includes Recommend Careers)	Student	US-04 / FR-11 to FR-14
UC-03	View Skill Gap Analysis	Student	US-05 / FR-15, FR-16
UC-04	View Personalized Roadmap (includes Generate Roadmap)	Student	US-06, US-08 / FR-17, FR-18
UC-05	View Course Recommendations	Student	US-07 / FR-19
UC-06	Chat with Career Bot (includes Answer Career Query)	Student	US-09 / FR-26 to FR-29
UC-07	Register / Login	Student, Admin	US-01 / FR-01 to FR-04
UC-08	View Analytics Dashboard	Student, Admin	US-12 / FR-21 to FR-25
UC-09	Manage Career Requirements	Admin	US-13 / FR-30
UC-10	Manage Courses	Admin	US-14 / FR-31
UC-11	Manage Users	Admin	FR-32

3. Use Case Diagram

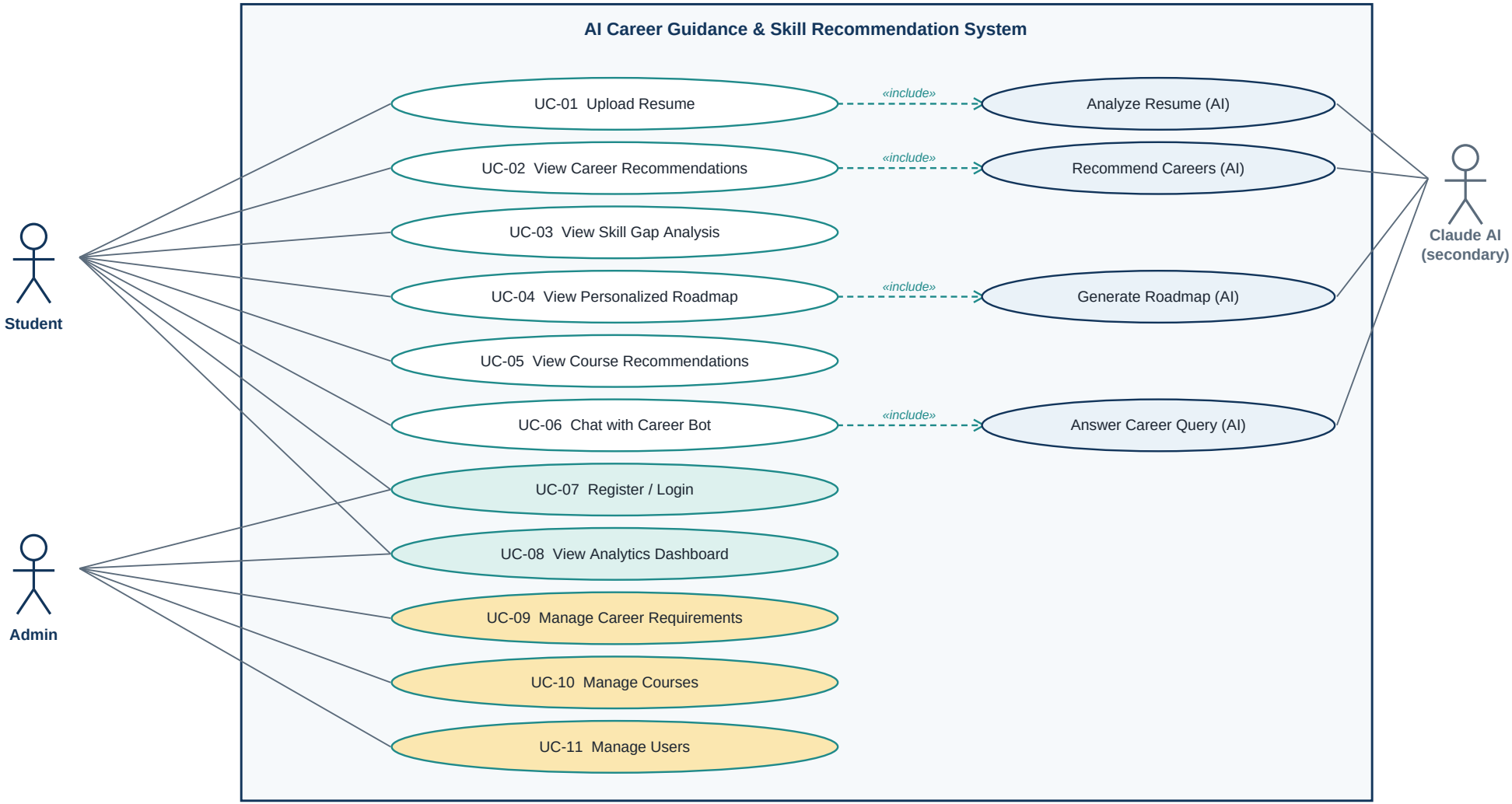

Figure 1: Use case diagram. Solid lines are associations; dashed arrows are «include» relationships to AI-supported use cases.

4. Use Case Descriptions (Narratives)

Use Case	**UC-01: Upload Resume**
Actors	Student (primary); Claude AI (secondary)
Description	Student uploads a resume; the system extracts text and Claude returns structured skills, education, projects and experience.
Preconditions	Student is logged in.
Main flow	1. Student opens the Resume page and selects a PDF or DOCX file. 2. System validates file type and size (max 5 MB). 3. System extracts raw text from the file. 4. System sends the text to Claude with an extraction prompt (include: Analyze Resume). 5. Claude returns structured JSON. 6. System stores the parsed profile and shows it to the student. 7. Student reviews and optionally edits the extracted data.
Alternate / exception flows	A1: Invalid file type or size, so the system shows an error and asks for another file. E1: Claude API fails or times out, so the system retries once, then shows a message and keeps the uploaded file for retry. E2: No text found (scanned image), so the system asks for a text-based resume.
Postconditions	Parsed profile saved and linked to the student; recommendations can be generated.

Use Case	**UC-02: View Career Recommendations**
Actors	Student; Claude AI
Description	System compares the student profile with stored career requirements and shows ranked careers with match scores and reasons.
Preconditions	A parsed resume profile exists.
Main flow	1. Student opens Recommendations. 2. System loads the profile and all career requirements. 3. System computes the weighted match score for each career. 4. Claude writes a short explanation for the top careers (include: Recommend Careers). 5. System displays careers ranked by score with matched skills. 6. Student selects a career as the target.
Alternate / exception flows	A1: No resume uploaded, so the system redirects to Upload Resume. E1: AI explanation fails, so scores are shown without explanations.
Postconditions	Recommendations stored; target career saved if selected.

Use Case	**UC-03: View Skill Gap Analysis**
Actors	Student
Description	System shows skills the student already has and the missing skills for the target career, with priority.
Preconditions	A target career is selected.
Main flow	1. Student opens Skill Gap. 2. System compares student skills with career requirements. 3. System lists matched skills and missing skills with priority (high, medium, low). 4. Student may change the target career to compare.
Alternate / exception flows	A1: Student changes career, so the gap is recalculated. A2: No gaps found, so the system shows a "ready for this role" message.
Postconditions	Skill gap report stored for analytics.

Use Case	**UC-04: View Personalized Roadmap**
Actors	Student; Claude AI
Description	System generates an ordered learning roadmap for the missing skills and tracks progress.
Preconditions	Skill gap report exists.
Main flow	1. Student opens Roadmap. 2. System sends missing skills and student level to Claude (include: Generate Roadmap). 3. Claude returns ordered steps with duration and goals. 4. System saves and shows the roadmap. 5. Student marks steps complete; progress updates.
Alternate / exception flows	E1: Claude fails, so the system builds a basic roadmap from stored skill order. A1: Student regenerates the roadmap.
Postconditions	Roadmap and progress saved.

Use Case	**UC-05: View Course Recommendations**
Actors	Student
Description	System suggests courses or resources for each missing skill.
Preconditions	Skill gap report exists.
Main flow	1. Student opens a roadmap step or the Courses page. 2. System fetches courses linked to the missing skills. 3. System shows courses with provider, level, cost and link. 4. Student opens a course link in a new tab.
Alternate / exception flows	A1: No course for a skill, so the system shows a general resource suggestion and notifies the admin.
Postconditions	No data change; click events may be logged.

Use Case	**UC-06: Chat with Career Bot**
Actors	Student; Claude AI
Description	Student asks career, skill, roadmap or preparation questions and receives AI answers using their profile as context.
Preconditions	Student is logged in.
Main flow	1. Student opens the chat and types a question. 2. System loads profile summary and recent messages. 3. System sends the prompt to Claude (include: Answer Career Query). 4. Claude returns an answer. 5. System shows and stores the reply.
Alternate / exception flows	A1: Off-topic question, so the bot politely redirects. E1: API error, so the system shows "try again" and keeps the user message.
Postconditions	Chat history saved.

Use Case	**UC-07: Register / Login**
Actors	Student, Admin
Description	User creates an account or signs in to access the system.
Preconditions	User has a valid email address.
Main flow	1. User opens the Login or Register page. 2. User enters credentials. 3. System validates input and checks or creates the account (password hashed). 4. System issues a JWT. 5. User is taken to the dashboard.
Alternate / exception flows	A1: Email already registered. E1: Wrong password, so an error appears; repeated failures are rate limited.
Postconditions	User session established.

Use Case	**UC-08: View Analytics Dashboard**
Actors	Student, Admin
Description	System displays aggregated charts of in-demand skills, popular careers and common skill gaps.
Preconditions	User is logged in.
Main flow	1. User opens the dashboard. 2. System requests aggregated data from the analytics module. 3. Analytics module computes statistics with Pandas. 4. System renders charts.
Alternate / exception flows	E1: Analytics unavailable, so cached data is shown with a notice.
Postconditions	None.

Use Case	**UC-09: Manage Career Requirements**
Actors	Admin
Description	Admin creates, edits or deletes careers and their required skills with weights.
Preconditions	Admin is logged in.
Main flow	1. Admin opens Careers management. 2. Admin adds or edits a career and its skill weights. 3. System validates data. 4. System saves changes.
Alternate / exception flows	A1: Delete a career in use, so the system asks for confirmation. E1: Validation fails, so errors are shown.
Postconditions	Career data updated and used in future recommendations.

Use Case	**UC-10: Manage Courses**
Actors	Admin
Description	Admin maintains the course catalogue linked to skills.
Preconditions	Admin is logged in.
Main flow	1. Admin opens Courses management. 2. Admin adds, edits or removes a course and links it to a skill. 3. System validates the URL and fields. 4. System saves changes.
Alternate / exception flows	E1: Invalid URL, so an error is shown.
Postconditions	Course catalogue updated.

Use Case	**UC-11: Manage Users**
Actors	Admin
Description	Admin views registered users and can deactivate accounts.
Preconditions	Admin is logged in.
Main flow	1. Admin opens Users. 2. System lists users with role and status. 3. Admin deactivates or reactivates an account. 4. System saves the change.
Alternate / exception flows	A1: Admin cannot deactivate own account.
Postconditions	User status updated.